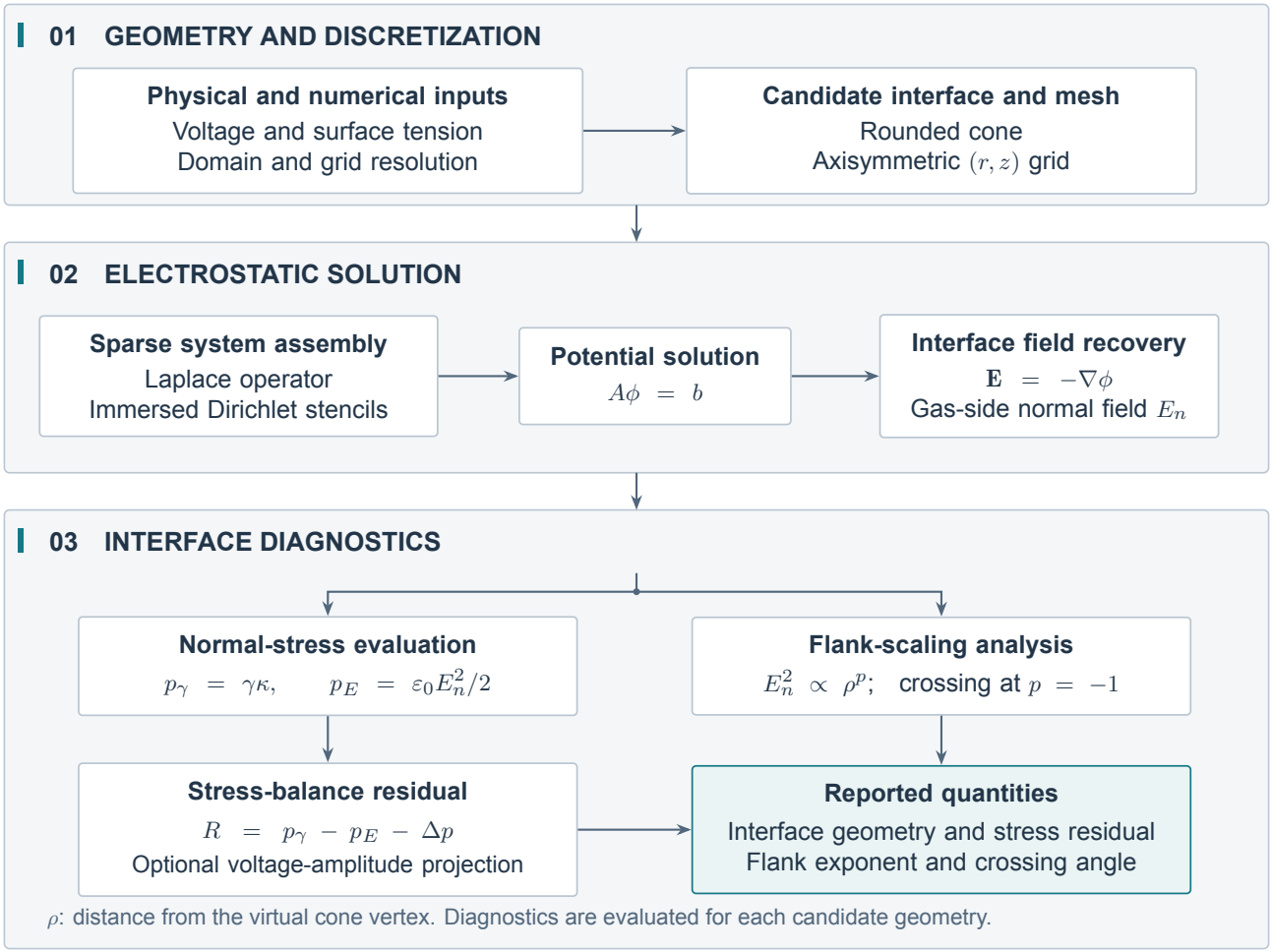
01 GEOMETRY AND DISCRETIZATION
Physical and numerical inputs
Voltage and surface tension
Domain and grid resolution
Candidate interface and mesh
Rounded cone
Axisymmetric (r, z) grid
02 ELECTROSTATIC SOLUTION
Sparse system assembly
Laplace operator
Immersed Dirichlet stencils
Potential solution
$A\phi = b$
Interface field recovery
$\mathbf{E} = -\nabla\phi$
Gas-side normal field E_n
03 INTERFACE DIAGNOSTICS
Normal-stress evaluation
$p_\gamma = \gamma\kappa, \quad p_E = \varepsilon_0 E_n^2/2$
Flank-scaling analysis
$E_n^2 \propto \rho^p$; crossing at $p = -1$
Stress-balance residual
$R = p_\gamma - p_E - \Delta p$
Optional voltage-amplitude projection
Reported quantities
Interface geometry and stress residual
Flank exponent and crossing angle
ρ: distance from the virtual cone vertex. Diagnostics are evaluated for each candidate geometry.